

ANKIT KUMAR

Department of Management Sciences (DoMS), Indian Institute of Technology Kanpur, UP – 208016, India
ankitk25@iitk.ac.in • [LinkedIn: in/ankit-kumar](https://www.linkedin.com/in/ankit-kumar) • [GitHub: Ankitiitk369](https://github.com/Ankitiitk369) • [Portfolio: ankit369portfolio](#)

Education

Indian Institute of Technology, Kanpur

Master of Technology (M.Tech), Department of Management Sciences (DoMS)

Cumulative Performance Index (CPI): **7.18 / 10.0**

Kanpur, India

July 2025 – Present

C.S.A. University of Agriculture and Technology

Bachelor of Technology (B.Tech) in Agricultural Engineering

Cumulative Performance Index (CPI): **8.25 / 10.0**

Kanpur, India

Aug 2021 – June 2025

*Awarded **Chancellor's Silver Medal***

Government Inter College

UP State Board: Class XII – 77.20% (2021) • Class X – 80.83% (2019)

Kadaura, Jalaun, UP, India

2018 – 2021

Technical Skills & Competencies

- **Languages & Core:** Python (Advanced), SQL (PostgreSQL, MySQL), C/C++, HTML, CSS, JavaScript, LaTeX
- **Machine Learning & Statistics:** Scikit-learn, XGBoost, LightGBM, CatBoost, Optuna, Feature Engineering, SMOTE, Hypothesis Testing (A/B Testing, ANOVA, Chi-Square), Statsmodels, Time Series (ARIMA, SARIMAX)
- **Deep Learning & Computer Vision:** PyTorch, TensorFlow, Keras, OpenCV, YOLOv8, EasyOCR, CNN, RNN, LSTM, Transfer Learning
- **Generative AI & Agentic Systems:** LangChain, LangGraph, LlamaIndex, CrewAI, Groq API, Pinecone, Qdrant, RA-GAS, Multimodal RAG, ReAct Framework, Prompt Engineering, LLMops
- **Analytics, BI & Tools:** Power BI (Advanced DAX), Tableau, Advanced Excel, FastAPI, Streamlit, Git, GitHub, Linux, Pytest, Docker
- **Relevant Coursework (IIT Kanpur):** Applied Machine Learning, Statistical Modelling for Business Analytics, Data Mining & Knowledge Discovery, Operations Research for Management, Computing, Supply Chain Management

Professional Experience

Tata Consultancy Services (TCS)

Remote / India

Data Science & AI Intern | Computer Vision & Machine Learning

May 2026 – July 2026

- Engineered an automated vehicle CO₂ emission tracking pipeline from real-time CCTV feeds to monitor urban carbon footprints for enterprise ESG reporting.
- Implemented **YOLOv8** object localization paired with **OpenCV** image preprocessing (Bilateral Filtering, CLAHE, Otsu) and **EasyOCR** with custom regex syntax correction.
- Trained dual-stage ensemble estimators: **Gradient Boosting ($R^2 = 0.996$, RMSE = 3.12 g/km)** and **Random Forest ($R^2 = 0.989$)**; delivered interactive **Streamlit** dashboard with **17-test Pytest** test suite.

Global Procurement Web Application System

Enterprise IT Project

Frontend & Asset Automation Contributor

Feb 2026 – Apr 2026

- Contributed to the Asset Receiving Automation module for an enterprise Global Procurement System, digitizing manual asset tracking workflows across 100+ inventory categories.
- Engineered responsive **Angular** user interfaces for capturing asset metadata, PO numbers, quantities, and storage locations with real-time client-side validation.
- Integrated dynamic form validation schemas, authentication workflows, and **Role-Based Access Control (RBAC)**, cutting asset ingestion latency by **40%**.

Key Projects: Generative AI & Autonomous Systems

MediMind AI: Clinical Decision Support Platform

Nov 2025 – Dec 2025

Stack: LangChain, Groq API (Llama-3), Pinecone, MiniLM, Streamlit

[\[GitHub Repo\]](#)

- Architected an end-to-end RAG clinical diagnostic platform using **LangChain**, **Groq API** (Llama-3), and **Pinecone** vector DB, slashing inference latency by **85%**.
- Integrated **MiniLM-L6-v2** embeddings with Cosine Similarity retrieval, securing **96.8% RAG faithfulness** and zero hallucinations with verified clinical citations.
- Engineered automated lab report ingestion via **PyPDF**, embedding 4 validated clinical risk calculators with sub-10ms emergency triage safety guardrails.

Key Projects: Machine Learning, Vision & Analytics (Continued)

Text-to-SQL & Agentic SQL Analytics System

Jan 2026 – Feb 2026

Stack: *LangGraph, Gemini 2.0 Flash, ReAct, MySQL, RAGAS, Plotly*

[\[GitHub Repo\]](#)

- Developed an autonomous Text-to-SQL analytics system utilizing **Gemini 2.0 Flash**, **LangChain**, and **MySQL** across a 6-table enterprise dataset (64,000+ orders).
- Architected an agentic **LangGraph ReAct** workflow with **SQLDatabaseToolkit** for autonomous schema inspection, query generation, and dynamic execution.
- Implemented a self-correcting query loop resolving syntax anomalies iteratively, achieving **1.00 Context Precision** on standardized **RAGAS** benchmarks.

Petrol Price Forecasting & Analytics System

Aug 2025 – Nov 2025

Academic Project (Guide: Prof. Faiz Hamid, IIT Kanpur) / Stack: *ARIMA, XGBoost, PyTorch LSTM, FastAPI* [\[GitHub Repo\]](#)

- Formulated a comparative time-series forecasting pipeline using **ARIMA(5,1,0)**, **XGBoost**, **PyTorch LSTM**, and Naive baselines for weekly commodity pricing.
- Engineered **15+ domain-specific temporal features** including 1–12 lag terms, 4-wk and 12-wk SMA/EMA, rolling volatility, and outlier filtering.
- Benchmarked models on 35% unseen test data; tuned **XGBoost** delivered **1.12% MAPE** and **67.99% directional accuracy**; deployed as REST API via **FastAPI**.

Credit Risk Modelling: Loan Eligibility Prediction

Aug 2025 – Nov 2025

Academic Project (Guide: Prof. Faiz Hamid, IIT Kanpur) / Stack: *Scikit-learn, XGBoost, SMOTE, GridSearchCV* [\[GitHub Repo\]](#)

- Constructed an enterprise credit scoring pipeline analyzing **51,336 customer records** with 79 financial attributes to predict loan default probabilities.
- Executed rigorous feature selection via **Chi-Square**, **ANOVA**, **Variance Inflation Factor (VIF)**, and tackled severe class imbalance via **SMOTE**.
- Evaluated Logistic Regression, Decision Trees, SVC, Random Forest, and **XGBoost**; optimized via **GridSearchCV**, achieving **78% out-of-sample accuracy**.

Solar Panel Defect Classification

Jan 2026 – May 2026

Academic Project (Guide: Prof. Veena Bansal, IIT Kanpur) / Stack: *TensorFlow, MobileNetV2, EfficientNetB0* [\[GitHub Repo\]](#)

- Trained a 6-class deep learning defect detection pipeline on **885 electroluminescence images** of photovoltaic cells using **TensorFlow** and transfer learning.
- Implemented data augmentation, Batch Normalization, Dropout (0.3), and L2 weight decay to combat small-sample overfitting and boost feature generalization.
- Benchmarked Custom CNN, **MobileNetV2**, and **Fine-tuned EfficientNetB0**, achieving **82.5% validation accuracy**; deployed inspection app via **Streamlit**.

Customer Shopping Behavior & Business Intelligence

Sep 2025 – Oct 2025

Stack: *PostgreSQL, Power BI, Advanced DAX, Pandas*

[\[GitHub Repo\]](#)

- Analyzed **3,900 consumer transactions** across 18 demographic and behavioral dimensions using **Pandas** and relational modeling.
- Formulated complex **PostgreSQL** queries utilizing **CTEs**, window ranking functions (**ROW_NUMBER**, **DENSE_RANK**), and multi-table joins.
- Engineered an executive **Power BI** dashboard with dynamic **DAX measures**, KPI cards, RFM segmentation, and drill-through cohort analysis.

Positions of Responsibility

- Alumni and Corporate Relations Coordinator, IIT Kanpur** (July 2025 – Present): Coordinate with corporate leadership and alumni network to drive campus engagement; organized 10+ departmental technical webinars and networking sessions.
- Campus Coordinator, Mimamsa 2025** (IISER Pune): Led institutional outreach across 15+ universities in northern India; supervised regional round operations.
- NSS Student Volunteer (2 Years)**: Spearheaded community education drives for girls' education advocacy and digital empowerment under '*Ek Bharat Shreshtha Bharat*'.

Scholastic Achievements & Honors

- GATE 2025**: Secured **98.48 percentile** in Agricultural Engineering, conducted by **Indian Institute of Technology, Roorkee**.
- Chancellor's Silver Medal Award**: Awarded the prestigious **Silver Medal** for rank 1st academic performance in B.Tech by the University Chancellor.
- GCI World Tokyo 2026**: Global Data Science & AI Program by **Matsuo-Iwasawa Laboratory, The University of Tokyo** (passed final evaluation).
- Certifications**: **Oracle Certified Data Science Professional** • **Oracle Certified AI Foundations** • **SEBI-Investor Certification (NISM)** • **British Airways Data Science** • **Deloitte Australia Analytics**.